

Table A2. Supplementary Geology Categories and Solar System Context

Category (1)	Definition (2)	Solar System Context (3)
Minerals	See Table A1	Minerals compose rocks and can be found in their pure form in rocks composed entirely of one mineral (e.g., anorthite and anorthosite, quartz and quartzite), or as large inclusions in rocks. On Earth's surface, minerals formed at high temperatures in the mantle (i.e., olivines, pyroxenes, feldspars) are broken down easily by atmospheric and surface water processes to form the basis of sedimentary rocks while stable minerals, like quartz, aren't. ^{Z16}
↔ Ferropicrinite	(Mg,Fe)O (also known as magnesio-wüstite), mineral category that encompasses periclase MgO and wüstite FeO	Makes up a portion of the lower mantle of Earth (alongside other lower mantle minerals like perovskite, CaTiO ₃). While stable at high pressures in the mantle, it is not stable in the low pressures found on the surface. ^{Z16} Due to this, no MgO or FeO albedo data is explicitly included in our database, but we include its definition here due to it being a potentially large mineralogical component of exotic upper mantle compositions (Putirka & Rarick 2019; Putirka & Xu 2021).
↔ Olivine	(Mg,Fe) ₂ SiO ₄ , mineral category that encompasses the endmembers forsterite Mg ₂ SiO ₄ to fayalite Fe ₂ SiO ₄	Olivine is one of the first minerals in a melt to crystallize out of a magma as a planet initially cools from formation, where due to its high density it often sinks. It can be present in its pure form as large crystals (phenocrysts) in rocks. On Earth's surface it often weathers quickly. ^{Z16} Alongside pyroxene, it represents the most abundant mineral in the upper mantle of Solar System rocky planets (Guimond et al. 2024). Olivine itself is also a large component of the material used to build planetesimals, and therefore has been found ubiquitously in the Solar System (Leone & Tanaka 2023).
↔ Pyroxene	XY(Si,Al) ₂ O ₆ , mineral category that encompasses enstatite MgSiO ₃ , orthopyroxenes (Mg,Fe) ₂ Si ₂ O ₆ , clinopyroxenes Ca(Mg,Fe)Si ₂ O ₆ , and augite	One of the most common minerals that crystallize first out of a magma. Alongside olivine, represents the most abundant mineral in the upper mantle of Solar System rocky planets (Guimond et al. 2024). ^{Z16}
↔ Silica	SiO ₂ , mineral group composed of multiple crystalline polymorphs (e.g., quartz, tridymite, cristobalite, coesite, opal)	Quartz is the most common polymorph of silica on Earth and comprises about a tenth of Earth's crust, though all polymorphs of silica can be found on Earth in small amounts (Richard Drees et al. 1989). Quartz is stable in the conditions of Earth's surface and therefore is the main component sand. Glass is amorphous silica. ^{Z16} Other polymorphs of silica can be found on the surfaces of planets; tridymite and cristobalite has been found on Mars and Moon (Seddio et al. 2015; Morris et al. 2016; Yen et al. 2021), and coesite has been found in impact craters. While not a main mineralogical component of the upper mantle in Solar System planets, it has been found to be a potentially large component of exotic upper mantle compositions (Putirka & Xu 2021).
↔ Feldspars	Aluminium-silicate mineral category that is composed of alkali feldspars (orthoclase, KAlSi ₃ O ₈ to albite, NaAlSi ₃ O ₈) and plagioclase feldspars (albite, NaAlSi ₃ O ₈ to anorthite, CaAl ₂ Si ₂ O ₈)	Feldspar minerals are the most abundant rock-forming mineral in Earth's crust, where the weathering of them forms clays. In primordial Earth and the Moon, plagioclase feldspars condensed out of calcium and aluminum enriched magma (which was depleted of iron and some silica due to olivine condensation) where, due to their low density, the crystallized material floated to the top and became a main component of the primary crust. ^{Z16}
↔ Pyrite	FeS ₂ , also known as Fool's Gold	Forms from iron and sulfur reacting together in an oxygen poor environment. It is the most abundant sulfide mineral on Earth. On Earth, it primarily forms in oxygen-poor underwater sediments via anaerobic microbes that strip oxygen from sulfate (SO ₄ ⁻²) ions to produce disulfide (S ₂ ⁻²) that reacts with reduced ferrous iron (Fe ⁺²) to form pyrite. Pyrite rapidly decomposes when exposed to the oxidizing environment of Earth's atmosphere. ^{Z16} Pyrite has been found on Mars (Vaniman et al. 2014), hinting that Mars once housed reducing environments.

Table A2 *continued*

Table A2 (continued)

Category (1)	Definition (2)	Solar System Context (3)
$\hookrightarrow$ Hematite	Fe_2O_3	Forms from iron bonding with oxygen, usually from the weathering of iron-rich materials in oxidizing conditions, or, formed from precipitation in aqueous environments. It is often found in the soil composition on Earth. When mixed with soils and clays, it can cause a red color; Mars has a thin layer of hematite on its surface causing its famous reddish color. ^{Z16} Hematite has been attributed to being a potential cause of the Fe^3 signal from Venus (Fegley et al. 1995), and has been recently discovered in the highlands of the Moon, despite its lack of a substantial atmosphere (Li et al. 2020).
Ultramafic Rocks	Category of rocks with chemical compositions that are the most magnesium-rich, silica-poor ($\text{SiO}_2 \leq 45$ wt%)	On Earth, ultramafic rocks were the first to condense out of the primordial magma ocean, forming the upper mantle. These rocks partially melt to form silica-rich magmas that then cool to form mafic rocks. ^{Z16}
$\hookrightarrow$ Peridotite	Rocks made from olivine and pyroxene minerals; by definition, is $> 40\%$ olivine	Peridotites are the main constituent of Earth's upper mantle, wherein their partial melts subsequently cool to form basalts. Models have also shown that peridotites are thought to be the main constituent of exoplanet upper mantles (Putirka & Xu 2021). Peridotites crystallize from magma early on, and can often be found as fragments in mafic rocks as 'xenoliths' that form when magma breaks off mantle rocks during its ascent to the surface. On Earth, large dislocated slabs of the lithosphere can be brought to observable crustal layers via tectonics and contain the full strata of basalts, gabbro, peridotite, and spinel (in that order of depth). ^{Z16} The relative abundance of olivine vs pyroxene in peridotite depends on the level of partial melting a rock has experienced, with example rocks in our database being dunite ($> 90\%$ olivine), harzburgite ($< 5\%$ clinopyroxene), and lherzolite (intermediate composition of clino- and ortho-pyroxene). ^{W14}
$\hookrightarrow$ Pyroxenite	Rocks made from olivines and pyroxenes; by definition, is $> 60\%$ pyroxene	While the olivines and pyroxenes minerals in Earth's upper mantle mostly produce peridotite, they produce a small percentage of pyroxenite as well. While it does not dominate any upper mantle in the Solar System, exotic mantle mineralogical compositions could produce abundant pyroxenites in the upper mantle (Putirka & Xu 2021).
Mafic Rocks	Broad category of rocks with chemical compositions that are relatively magnesium (Mg) rich ($\text{SiO}_2 < 63$ wt%)	On Earth, mafic rocks form from the cooling of partial melts of ultramafic mantle material, where it forms fresh crustal material at mid-ocean ridges. A majority of Solar System rocky body surfaces are composed of mafic rocks. ^{Z16}

Table A2 continued

Table A2 (continued)

Category (1)	Definition (2)	Solar System Context (3)
↔ Basalts (TAS)	Rocks formed from partial melts of ultramafic material, which contain relatively low SiO ₂ (46-52 wt%) and low Na ₂ O+K ₂ O (<5 wt%) when compared to other TAS rocks. Basalts have a mineralogical composition comprised mostly of feldspars and pyroxenes, and its intrusive variant is gabbro	Basalts are the most common igneous rock on the surface of Earth and are the main component of Earth's oceanic crust, where fresh crust is created continuously at mid-ocean ridges. They are the most ubiquitous surface component of other Solar System terrestrial planets and moons (Mercury, Venus, southern highlands of Mars, dark lowland regions of the Moon, volcanic eruptions of Io) despite different internal structures and large-scale geologic processes in these objects. Most non-exotic silicate mantles are predicted produce basaltic magmas; where, in general, partial melts of ultramafic mantle material will cool and produce a basalt (First et al. 2025). ^{Z16} The two main sub-categories are alkali basalts, which have higher Na ₂ O+K ₂ O content, and tholeiitic basalts, which are more SiO ₂ saturated. Tholeiitic basalts are the most common form of basalt on Earth, being the form of basalts that emerge at mid-ocean ridges (they are also the most common form of basalt on the Moon) and represent more shallow mantle melting, while alkali basalts form in regions of low tectonic activity and represent more deeper mantle melting (like volcanic islands and continental rifts). ^{W14} Basalts are usually dark-colored due to augite or pyroxene minerals, but can be lighter due to feldspar minerals. It has been shown that basalt's mid-infrared absorption feature cannot be modeled by mixing individual mineral absorption features, highlighting the necessity of lab data (Hu et al. 2012; Ehlmann & Mustard 2012).
↔ Trachybasalt (TAS)	Similar SiO ₂ content to basalts, but with a higher content of Na ₂ O+K ₂ O (5-7.3 wt%)	It is found in Earth on some ocean islands, continental rift volcanoes, ^{W14} and has been found in the Gale Crater on Mars (Edwards et al. 2017).
↔ Basaltic Andesite (TAS)	Slightly higher SiO ₂ content to basalts (52-57 wt%), but with a similar Na ₂ O+K ₂ O content	Found in volcanic arcs (a belt of volcanoes above a subducting plate) on Earth, and have also been found on Mars (Ruff & Christensen 2007; Mullen & McCallum 2013).
↔ Tephrite (TAS)	Distinguished from basalt by a lower SiO ₂ content (41-49 wt%) and higher Na ₂ O+K ₂ O content (3-9.4 wt%)	Found in specific regions of Earth and Mars where local magma was enriched in alkaline material. ^{W14}
↔ Phonolite (TAS)	Higher SiO ₂ content than basalts (52.5-61 wt%), and is the TAS category with the highest amount of Na ₂ O+K ₂ O content (> 12 wt%)	Rare on Earth, where their formation requires a very low degree of partial melting such that alkali-rich rocks melt but silicon-rich rocks don't. ^{W14}
↔ Gabbro	Is the intrusive variant of basalt, which is more coarse grained	When compared to basalts, which cool rapidly at the surface of fresh oceanic crust due to the magma's interaction with water, gabbro forms in the denser sub-surface portion of oceanic crust on Earth where it can cool slowly (and therefore form larger grains). ^{Z16}
↔ Anorthosite	Rock composed of plagioclase feldspar minerals	Anorthosite is a large component of the Moon's surface (bright areas). When plagioclase feldspars condense into anorthosite, they are less dense than the surrounding magma and float to the top, creating a bright, primary crust. ^{Z16}
Intermediate Rocks		
↔ Andesite (TAS)	Distinguished from basalt by higher SiO ₂ content (57-73 wt%) with an identical Na ₂ O+K ₂ O content; its intrusive variant is diorite	Basalts and andesites are often found together, though basalts are more indicative of oceanic crust and ocean floor spreading, while andesite is commonly found in subduction zones (particularly in island arcs) and has been considered a hallmark of Earth's plate tectonics (Tatsumi et al. 2015). Andesitic magma forms from the same mechanism that produces granitic and rhyolitic magma (see Figure 4), but rapidly cools on the surface (and therefore is not coarsely textured). Andesite has also been found in the crust of Mars, and theorized to be present on Venus (Pavri et al. 1992; McSween et al. 2009).

Table A2 continued

Table A2 (continued)

Category (1)	Definition (2)	Solar System Context (3)
↔ Trachyte (TAS)	Higher SiO ₂ content than basalts (57-69 wt%) with a large amount of Na ₂ O+K ₂ O content (> 7 wt%)	Found in specific regions of Earth where alkaline magma has cooled enough (where refractory materials condense and remove themselves from the melt) that the remaining magma has a composition similar to alkali feldspars. Rarely, on Earth, do more alkali-rich magmas form (that generate phonolite). ^{W14} It has been found in the Gale Crater on Mars (Sautter et al. 2016).
Felsic Rocks	Broad category composing of relatively silicon (Si) rich (SiO ₂ ≥ 63 wt%) material	On Earth, felsic rocks form from partially melted mafic parent material, typically at subduction zones, where the partial melts are enriched in silica. Earth has the highest percentage of felsic material on the surface when compared to other Solar System planets, due to processes like plate tectonics gradually producing large amounts of continental crust (from repeated processing of oceanic crust) over geologic timescales. ^{Z16}
↔ Granite	Rocks formed from the partial melts of mafic material, which contain very high SiO ₂ (≥ 70 wt%) and low MgO (≤ 1 wt%), with a mineralogical composition comprised mostly of quartz and feldspars; extrusive variant is rhyolite	Granite is the main component of Earth’s continental crust, where its parent, silica-rich magma is typically formed on Earth by the partial melting of mafic (basaltic) oceanic crust at subduction zones. The silica-rich magma then ascends and cools slowly, forming a coarse grained rock that makes up the continental crust. ^{Z16} In the Solar System, a widespread presence of granite is unique to Earth due to plate tectonics, though minimal amounts have been found on Mars (Bonin 2012).
↔ Rhyolite (TAS)	Highest amount of SiO ₂ content (> 69%) with large range of Na ₂ O+K ₂ O content (0-13 wt%); intrusive variant is granite	Found wherever silicon-rich lava rapidly cools, and is a common component of volcanic ash (where the silica-rich, viscous nature of rhyolitic magma often results in explosive volcanic eruptions with pyroclastic rocks, on Earth). Forms via the same mechanism as granite, except the parent rhyolitic lava cools rapidly on the surface (forming a fine-grained rock, in lieu of slowly cooled, coarse-grained granite). Obsidian is a rhyolitic glass, pumice is often rhyolitic, and rhyolitic tuffs are a common structure on Earth’s surface. ^{Z16} Rhyolitic-like lava has been theorized on Mars due to the discovery of tridymite (Morris et al. 2016; Yen et al. 2021).
Others		
↔ Clays	Category of sedimentary, soil materials (fine-grained) that forms from clay minerals (hydrous aluminum phyllosilicates)	Clays form in the presence of water. While they are very abundant on Earth ^{Z16} , they are relatively rare on other bodies (though clays have been found on Mars (Du et al. 2023) and Ceres (Rivkin et al. 2006)), indicating past or current surface water.
↔ Salts	Category of minerals formed from the combination of a positive and negative ion, that commonly precipitate from surface water	The most common salts in Earth’s ocean are NaCl (table salt) and MgSO ₄ (epsom salt) (Loganathan et al. 2017). Solid NaCl has been found in salt deposits on Earth and Mars (indicative of dried-up salt beds) (Bramble & Hand 2024), the sub-surface of Europa (Trumbo et al. 2019), and potential salt glacier have been proposed to exist in regions of Mercury (Rodriguez et al. 2023). MgSO ₄ similarly has been found on Earth and Mars (Vaniman et al. 2004).

NOTE—Geological categories used to sort the POSEIDON surface albedo database in Table A4. As in Table A1, this table is not extensive and was generated to help guide future considerations for exogeology studies and foster collaboration between geologists and exoplanet scientists. Basalt and Granite in boldface due to their extensive use in this paper. (1) Category name. The five large categories are Minerals, Mafic Rocks, Intermediate Rocks, Felsic Rocks, and Others. We take our ultramafic, mafic, and felsic definitions from Paragas et al. (2025). Instead of having a distinct intermediate category, as in Paragas et al. (2025), we only label rock categories as intermediate if their TAS-defined SiO₂ range falls between our mafic and felsic distinguishing SiO₂ percentage (63%). If a sub-category has (TAS) beside it, it means that it is a specific rock category in the Total Alkali-Silica (TAS) classification diagram where the ranges of SiO₂ and alkali metal oxide content for each category are taken from Le Maitre et al. (2005). We note that that the database presented in this work does include representative lab data for the following TAS fields: foidite, microbasalt, basanite, phonotephryte, tephryphonolite, basaltic trachyandesite, trachyandensite, trachydacite, and dacite. (2) Definition of the category. (3) Solar-System Context of the category. ^{W14} indicates that information preceding the superscript was summarized from Winter (2014). ^{Z16} indicates that information preceding the superscript was summarized from Zalasiewicz (2016).

REFERENCES

- Adams Redai, J., Wogan, N., Wallack, N. L., et al. 2025, *AJ*, 170, 219, doi: [10.3847/1538-3881/adee92](https://doi.org/10.3847/1538-3881/adee92)
- Allen, N. H., Espinoza, N., Diamond-Lowe, H., et al. 2025, *AJ*, 170, 240, doi: [10.3847/1538-3881/adfc51](https://doi.org/10.3847/1538-3881/adfc51)
- August, P. C., Buchhave, L. A., Diamond-Lowe, H., et al. 2025, *A&A*, 695, A171, doi: [10.1051/0004-6361/202452611](https://doi.org/10.1051/0004-6361/202452611)
- Baldrige, A. M., Hook, S. J., Grove, C. I., & Rivera, G. 2009, *Remote Sensing of Environment*, 113, 711, doi: [10.1016/j.rse.2008.11.007](https://doi.org/10.1016/j.rse.2008.11.007)
- Barsukov, V. L., Volkov, V. P., & Khodakovskii, I. L. 1982, *Lunar and Planetary Science Conference Proceedings*, 87, 3, doi: [10.1029/JB087iS01p000A3](https://doi.org/10.1029/JB087iS01p000A3)
- Batalha, N. E., Marley, M. S., Lewis, N. K., & Fortney, J. J. 2019, *ApJ*, 878, 70, doi: [10.3847/1538-4357/ab1b51](https://doi.org/10.3847/1538-4357/ab1b51)
- Batalha, N. E., Mandell, A., Pontoppidan, K., et al. 2017, *PASP*, 129, 064501, doi: [10.1088/1538-3873/aa65b0](https://doi.org/10.1088/1538-3873/aa65b0)
- Bédard, J. H. 2018, *Geoscience Frontiers*, 9, 19, doi: [10.1016/j.gsf.2017.01.005](https://doi.org/10.1016/j.gsf.2017.01.005)
- Bello-Arufe, A., Damiano, M., Bennett, K. A., et al. 2025, *ApJL*, 980, L26, doi: [10.3847/2041-8213/adaf22](https://doi.org/10.3847/2041-8213/adaf22)
- Benneke, B., & Seager, S. 2012, *ApJ*, 753, 100, doi: [10.1088/0004-637X/753/2/100](https://doi.org/10.1088/0004-637X/753/2/100)
- Bonin, B. 2012, *Lithos*, 153, 3, doi: [10.1016/j.lithos.2012.04.007](https://doi.org/10.1016/j.lithos.2012.04.007)
- Bramble, M. S., & Hand, K. P. 2024, *Scientific Reports*, 14, 5503, doi: [10.1038/s41598-024-55979-6](https://doi.org/10.1038/s41598-024-55979-6)
- Breuer, D., Hamilton, C. W., & Khurana, K. 2022, *Elements*, 18, 385, doi: [10.2138/gselements.18.6.385](https://doi.org/10.2138/gselements.18.6.385)
- Brugman, K., Phillips, M. G., & Till, C. B. 2021, *Journal of Geophysical Research (Planets)*, 126, e06731, doi: [10.1029/2020JE006731](https://doi.org/10.1029/2020JE006731)
- Cadieux, C., Doyon, R., MacDonald, R. J., et al. 2024, *ApJL*, 970, L2, doi: [10.3847/2041-8213/ad5afa](https://doi.org/10.3847/2041-8213/ad5afa)
- Cambianica, P., Simioni, E., Cremonese, G., et al. 2024, *Planetary and Space Science*, 253, 105983, doi: <https://doi.org/10.1016/j.pss.2024.105983>
- Cambioni, S., Weiss, B. P., Asphaug, E., et al. 2025, *A&A*, 696, A174, doi: [10.1051/0004-6361/202450128](https://doi.org/10.1051/0004-6361/202450128)
- Carrión-González, Ó., García Muñoz, A., Cabrera, J., et al. 2020, *A&A*, 640, A136, doi: [10.1051/0004-6361/202038101](https://doi.org/10.1051/0004-6361/202038101)
- Carter, L. M., Gilmore, M. S., Ghail, R. C., et al. 2023, *SSRv*, 219, 85, doi: [10.1007/s11214-023-01033-2](https://doi.org/10.1007/s11214-023-01033-2)
- Chao, K.-H., deGraffenried, R., Lach, M., et al. 2021, *Chemie der Erde / Geochemistry*, 81, 125735, doi: [10.1016/j.chemer.2020.125735](https://doi.org/10.1016/j.chemer.2020.125735)
- Cheek, L. C., Pieters, C. M., Dyar, M. D., & Milam, K. A. 2009, in *40th Annual Lunar and Planetary Science Conference*, Lunar and Planetary Science Conference, 1928
- Condie, K. C. 2022, in *Earth as an Evolving Planetary System (Fourth Edition)*, fourth edition edn., ed. K. C. Condie (Academic Press), 305–352, doi: <https://doi.org/10.1016/B978-0-12-819914-5.00009-3>
- Cottrell, E., Canil, D., Langmuir, C., Evans, K. A., & Gaillard, F. 2025, *Nature Reviews Earth and Environment*, 6, 728, doi: [10.1038/s43017-025-00735-1](https://doi.org/10.1038/s43017-025-00735-1)
- Coulombe, L.-P., Benneke, B., Challener, R., et al. 2023, *Nature*, 620, 292, doi: [10.1038/s41586-023-06230-1](https://doi.org/10.1038/s41586-023-06230-1)
- Cowan, N. B., Lustig-Yaeger, J., Hu, R., Mayorga, L. C., & Robinson, T. D. 2025, *arXiv e-prints*, arXiv:2507.03071, doi: [10.48550/arXiv.2507.03071](https://doi.org/10.48550/arXiv.2507.03071)
- Coy, B. P., Ih, J., Kite, E. S., et al. 2025, *ApJ*, 987, 22, doi: [10.3847/1538-4357/add3f7](https://doi.org/10.3847/1538-4357/add3f7)
- Dai, L., Titov, D. V., Shao, W. D., et al. 2025, *SSRv*, 221, 51, doi: [10.1007/s11214-025-01176-4](https://doi.org/10.1007/s11214-025-01176-4)
- Damiano, M., Bello-Arufe, A., Yang, J., & Hu, R. 2024, *ApJL*, 968, L22, doi: [10.3847/2041-8213/ad5204](https://doi.org/10.3847/2041-8213/ad5204)
- Damiano, M., & Hu, R. 2020, *AJ*, 159, 175, doi: [10.3847/1538-3881/ab79a5](https://doi.org/10.3847/1538-3881/ab79a5)
- Deng, J., Du, Z., Karki, B. B., Ghosh, D. B., & Lee, K. K. M. 2020, *Nature Communications*, 11, 2007, doi: [10.1038/s41467-020-15757-0](https://doi.org/10.1038/s41467-020-15757-0)
- Domingue, D. L., Koehn, P. L., Killen, R. M., et al. 2007, *SSRv*, 131, 161, doi: [10.1007/s11214-007-9260-9](https://doi.org/10.1007/s11214-007-9260-9)
- Dorschner, J., Begemann, B., Henning, T., Jaeger, C., & Mutschke, H. 1995, *A&A*, 300, 503
- Doyle, A. E., Young, E. D., Klein, B., Zuckerman, B., & Schlichting, H. E. 2019, *Science*, 366, 356, doi: [10.1126/science.aax3901](https://doi.org/10.1126/science.aax3901)
- Drant, T., Tian, M., Carrasco, N., & Heng, K. 2025, *A&A*, 698, A76, doi: [10.1051/0004-6361/202452016](https://doi.org/10.1051/0004-6361/202452016)
- Driscoll, P. E., & Barnes, R. 2015, *Astrobiology*, 15, 739, doi: [10.1089/ast.2015.1325](https://doi.org/10.1089/ast.2015.1325)
- Du, P., Yuan, P., Liu, J., & Ye, B. 2023, *Earth Science Reviews*, 243, 104491, doi: [10.1016/j.earscirev.2023.104491](https://doi.org/10.1016/j.earscirev.2023.104491)
- Ducrot, E., Lagage, P.-O., Min, M., et al. 2025, *Nature Astronomy*, 9, 358, doi: [10.1038/s41550-024-02428-z](https://doi.org/10.1038/s41550-024-02428-z)
- East, M., Miller, P., Brennan, S., & Lunde, E. 2024, in *Society of Photo-Optical Instrumentation Engineers (SPIE) Conference Series*, Vol. 13092, *Space Telescopes and Instrumentation 2024: Optical, Infrared, and Millimeter Wave*, ed. L. E. Coyle, S. Matsuura, & M. D. Perrin, 130921Q, doi: [10.1117/12.3020201](https://doi.org/10.1117/12.3020201)

- Edwards, P. H., Bridges, J. C., Wiens, R., et al. 2017, *M&PS*, 52, 2931, doi: [10.1111/maps.12953](https://doi.org/10.1111/maps.12953)
- Egan, W. G., & Hilgeman, T. 1975, *AJ*, 80, 587, doi: [10.1086/111782](https://doi.org/10.1086/111782)
- Ehlmann, B. L., & Mustard, J. F. 2012, *Geophys. Res. Lett.*, 39, L11202, doi: [10.1029/2012GL051594](https://doi.org/10.1029/2012GL051594)
- Elkins-Tanton, L. T., Hess, P. C., & Parmentier, E. M. 2005, *Journal of Geophysical Research (Planets)*, 110, E12S01, doi: [10.1029/2005JE002480](https://doi.org/10.1029/2005JE002480)
- Espinoza, N., Allen, N. H., Glidden, A., et al. 2025, *ApJL*, 990, L52, doi: [10.3847/2041-8213/adf42e](https://doi.org/10.3847/2041-8213/adf42e)
- Fabian, D., Henning, T., Jäger, C., et al. 2001, *A&A*, 378, 228, doi: [10.1051/0004-6361:20011196](https://doi.org/10.1051/0004-6361:20011196)
- Fegley, Jr., B., Klingelhofer, G., Brackett, R. A., et al. 1995, *Icarus*, 118, 373, doi: [10.1006/icar.1995.1197](https://doi.org/10.1006/icar.1995.1197)
- Feng, Y. K., Robinson, T. D., Fortney, J. J., et al. 2018, *AJ*, 155, 200, doi: [10.3847/1538-3881/aab95c](https://doi.org/10.3847/1538-3881/aab95c)
- Feroz, F., Hobson, M. P., & Bridges, M. 2009, *MNRAS*, 398, 1601, doi: [10.1111/j.1365-2966.2009.14548.x](https://doi.org/10.1111/j.1365-2966.2009.14548.x)
- First, E. C., Mishra, I., Gazel, E., et al. 2025, *Nature Astronomy*, 9, 370, doi: [10.1038/s41550-024-02412-7](https://doi.org/10.1038/s41550-024-02412-7)
- Fortin, M.-A., Gazel, E., Kaltenegger, L., & Holycross, M. E. 2022, *MNRAS*, 516, 4569, doi: [10.1093/mnras/stac2198](https://doi.org/10.1093/mnras/stac2198)
- Fortin, M.-A., Gazel, E., Williams, D. B., et al. 2024, *ApJL*, 974, L7, doi: [10.3847/2041-8213/ad7d89](https://doi.org/10.3847/2041-8213/ad7d89)
- Fortune, M., Gibson, N. P., Diamond-Lowe, H., et al. 2025, *A&A*, 701, A25, doi: [10.1051/0004-6361/202554198](https://doi.org/10.1051/0004-6361/202554198)
- Gaillard, F., Bouhifd, M. A., Füri, E., et al. 2021, *SSRv*, 217, 22, doi: [10.1007/s11214-021-00802-1](https://doi.org/10.1007/s11214-021-00802-1)
- Gazel, E., Hayes, J. L., Hoernle, K., et al. 2015, *Nature Geoscience*, 8, 321, doi: [10.1038/ngeo2392](https://doi.org/10.1038/ngeo2392)
- Gkouvelis, L. 2025, *PSJ*, 6, 110, doi: [10.3847/PSJ/adcba8](https://doi.org/10.3847/PSJ/adcba8)
- Glidden, A., Ranjan, S., Seager, S., et al. 2025, *ApJL*, 990, L53, doi: [10.3847/2041-8213/adf62e](https://doi.org/10.3847/2041-8213/adf62e)
- Gomez Barrientos, J., Kaltenegger, L., & MacDonald, R. J. 2023a, *MNRAS*, 524, L5, doi: [10.1093/mnras/slad056](https://doi.org/10.1093/mnras/slad056)
- Gomez Barrientos, J., MacDonald, R. J., Lewis, N. K., & Kaltenegger, L. 2023b, *ApJ*, 946, 96, doi: [10.3847/1538-4357/aca5f9](https://doi.org/10.3847/1538-4357/aca5f9)
- Goodis Gordon, K. E., Karalidi, T., Bott, K. M., et al. 2025, *ApJ*, 983, 168, doi: [10.3847/1538-4357/adc09c](https://doi.org/10.3847/1538-4357/adc09c)
- Gordon, I. E., Rothman, L. S., Hargreaves, R. J., et al. 2022, *JQSRT*, 277, 107949, doi: [10.1016/j.jqsrt.2021.107949](https://doi.org/10.1016/j.jqsrt.2021.107949)
- Grant, D., Lewis, N. K., Wakeford, H. R., et al. 2023, *ApJL*, 956, L32, doi: [10.3847/2041-8213/acfc3b10.3847/2041-8213/acfdab](https://doi.org/10.3847/2041-8213/acfc3b10.3847/2041-8213/acfdab)
- Greene, T. P., Bell, T. J., Ducrot, E., et al. 2023, *Nature*, 618, 39, doi: [10.1038/s41586-023-05951-7](https://doi.org/10.1038/s41586-023-05951-7)
- Gu, L., Fan, S., Li, J., et al. 2021, *AJ*, 161, 122, doi: [10.3847/1538-3881/abd54a](https://doi.org/10.3847/1538-3881/abd54a)
- Guimond, C. M., Shorttle, O., Jordan, S., & Rudge, J. F. 2023, *MNRAS*, 525, 3703, doi: [10.1093/mnras/stad2486](https://doi.org/10.1093/mnras/stad2486)
- Guimond, C. M., Wang, H., Seidler, F., et al. 2024, *Reviews in Mineralogy and Geochemistry*, 90, 259, doi: [10.2138/rmg.2024.90.08](https://doi.org/10.2138/rmg.2024.90.08)
- Hale, G. M., & Querry, M. R. 1973, *ApOpt*, 12, 555, doi: [10.1364/AO.12.000555](https://doi.org/10.1364/AO.12.000555)
- Hammond, M., Guimond, C. M., Lichtenberg, T., et al. 2025, *ApJL*, 978, L40, doi: [10.3847/2041-8213/ada0bc](https://doi.org/10.3847/2041-8213/ada0bc)
- Hapke, B. 1981, *J. Geophys. Res.*, 86, 3039, doi: [10.1029/JB086iB04p03039](https://doi.org/10.1029/JB086iB04p03039)
- . 2002, *Icarus*, 157, 523, doi: [10.1006/icar.2002.6853](https://doi.org/10.1006/icar.2002.6853)
- Hapke, B. 2012, *Theory of Reflectance and Emittance Spectroscopy*, 2nd edn. (Cambridge University Press)
- Hazen, R. M., Downs, R. T., Morrison, S. M., et al. 2023, *Journal of Geophysical Research: Planets*, 128, e2023JE007865, doi: <https://doi.org/10.1029/2023JE007865>
- He, C., Radke, M., Moran, S. E., et al. 2024, *Nature Astronomy*, 8, 182, doi: [10.1038/s41550-023-02140-4](https://doi.org/10.1038/s41550-023-02140-4)
- Head, J. W., Chapman, C. R., Domingue, D. L., et al. 2007, *SSRv*, 131, 41, doi: [10.1007/s11214-007-9263-6](https://doi.org/10.1007/s11214-007-9263-6)
- Heng, K., Morris, B. M., & Kitzmann, D. 2021, *Nature Astronomy*, 5, 1001, doi: [10.1038/s41550-021-01444-7](https://doi.org/10.1038/s41550-021-01444-7)
- Herbort, O., & Sereinig, L. 2025, *A&A*, 699, A67, doi: [10.1051/0004-6361/202453069](https://doi.org/10.1051/0004-6361/202453069)
- Hinkel, N. R., Timmes, F. X., Young, P. A., Pagano, M. D., & Turnbull, M. C. 2014, *AJ*, 148, 54, doi: [10.1088/0004-6256/148/3/54](https://doi.org/10.1088/0004-6256/148/3/54)
- Hinkel, N. R., & Unterborn, C. T. 2018, *ApJ*, 853, 83, doi: [10.3847/1538-4357/aaa5b4](https://doi.org/10.3847/1538-4357/aaa5b4)
- Hinkel, N. R., Youngblood, A., & Soares-Furtado, M. 2024, *Reviews in Mineralogy and Geochemistry*, 90, 1, doi: [10.2138/rmg.2024.90.01](https://doi.org/10.2138/rmg.2024.90.01)
- Hohm, U. 1994, *Chemical Physics*, 179, 533, doi: [10.1016/0301-0104\(94\)87028-4](https://doi.org/10.1016/0301-0104(94)87028-4)
- Holmberg, M., Diamond-Lowe, H., Mendonça, J. M., et al. 2026, *The Astronomical Journal*, 171, 251, doi: [10.3847/1538-3881/ae4c45](https://doi.org/10.3847/1538-3881/ae4c45)
- Hu, R., Ehlmann, B. L., & Seager, S. 2012, *ApJ*, 752, 7, doi: [10.1088/0004-637X/752/1/7](https://doi.org/10.1088/0004-637X/752/1/7)
- Hu, R., Bello-Arufe, A., Zhang, M., et al. 2024, *Nature*, 630, 609, doi: [10.1038/s41586-024-07432-x](https://doi.org/10.1038/s41586-024-07432-x)
- Hu, R., Min, M., Millar-Blanchaer, M., et al. 2025, *arXiv e-prints*, arXiv:2509.16798, doi: [10.48550/arXiv.2509.16798](https://doi.org/10.48550/arXiv.2509.16798)

- Hunten, D. M. 1993, *Science*, 259, 915, doi: [10.1126/science.259.5097.915](https://doi.org/10.1126/science.259.5097.915)
- Ih, J., Kempton, E. M. R., Whittaker, E. A., & Lessard, M. 2023, *ApJL*, 952, L4, doi: [10.3847/2041-8213/ace03b](https://doi.org/10.3847/2041-8213/ace03b)
- Ito, Y., Ikoma, M., Kawahara, H., et al. 2015, *ApJ*, 801, 144, doi: [10.1088/0004-637X/801/2/144](https://doi.org/10.1088/0004-637X/801/2/144)
- Jones, K. D., Morris, B. M., & Heng, K. 2025, *A&A*, 694, A288, doi: [10.1051/0004-6361/202451115](https://doi.org/10.1051/0004-6361/202451115)
- Kaltenegger, L., MacDonald, R. J., Kozakis, T., et al. 2020, *ApJL*, 901, L1, doi: [10.3847/2041-8213/aba9d3](https://doi.org/10.3847/2041-8213/aba9d3)
- Karman, T., Gordon, I. E., van der Avoird, A., et al. 2019, *Icarus*, 328, 160, doi: [10.1016/j.icarus.2019.02.034](https://doi.org/10.1016/j.icarus.2019.02.034)
- Khare, B. N., Sagan, C., Arakawa, E. T., et al. 1984, *Icarus*, 60, 127, doi: [10.1016/0019-1035\(84\)90142-8](https://doi.org/10.1016/0019-1035(84)90142-8)
- Kite, E. S., Fegley, Jr., B., Schaefer, L., & Gaidos, E. 2016, *ApJ*, 828, 80, doi: [10.3847/0004-637X/828/2/80](https://doi.org/10.3847/0004-637X/828/2/80)
- Koike, C., Kaito, C., Yamamoto, T., et al. 1995, *Icarus*, 114, 203, doi: [10.1006/icar.1995.1055](https://doi.org/10.1006/icar.1995.1055)
- Kokaly, R. F., Clark, R. N., Swayze, G. A., et al. 2017, *USGS Spectral Library Version 7*, U.S. Geological Survey, Crustal Geophysics and Geochemistry Science Center, USGS Data Series, Report: iv, 61 p.; Dataset; Data Release, doi: [10.3133/ds1035](https://doi.org/10.3133/ds1035)
- Koll, D. D. B., Malik, M., Mansfield, M., et al. 2019, *ApJ*, 886, 140, doi: [10.3847/1538-4357/ab4c91](https://doi.org/10.3847/1538-4357/ab4c91)
- Kreidberg, L., & Stevenson, K. B. 2025, *Proceedings of the National Academy of Science*, 122, e2416190122, doi: [10.1073/pnas.2416190122](https://doi.org/10.1073/pnas.2416190122)
- Krissansen-Totton, J., Ulses, A. G., Frissell, M., et al. 2025, *arXiv e-prints*, arXiv:2507.14771, doi: [10.48550/arXiv.2507.14771](https://doi.org/10.48550/arXiv.2507.14771)
- Lammer, H., Zerkle, A. L., Gebauer, S., et al. 2018, *A&A Rv*, 26, 2, doi: [10.1007/s00159-018-0108-y](https://doi.org/10.1007/s00159-018-0108-y)
- Le Maitre, R. W., Streckeisen, A., Zanettin, B., et al. 2005, *Igneous Rocks: A Classification and Glossary of Terms*
- Leone, G., & Tanaka, H. K. 2023, *iScience*, 26, 107160, doi: <https://doi.org/10.1016/j.isci.2023.107160>
- Lewis, K. W., & Aharonson, O. 2014, *Journal of Geophysical Research (Planets)*, 119, 1432, doi: [10.1002/2013JE004404](https://doi.org/10.1002/2013JE004404)
- Li, G., Gordon, I. E., Rothman, L. S., et al. 2015, *ApJS*, 216, 15, doi: [10.1088/0067-0049/216/1/15](https://doi.org/10.1088/0067-0049/216/1/15)
- Li, S., Lucey, P. G., Fraeman, A. A., et al. 2020, *Science Advances*, 6, eaba1940, doi: [10.1126/sciadv.aba1940](https://doi.org/10.1126/sciadv.aba1940)
- Liang, W., Broquet, A., Andrews-Hanna, J. C., et al. 2024, *Nature Geoscience*, 17, 361, doi: [10.1038/s41561-024-01408-2](https://doi.org/10.1038/s41561-024-01408-2)
- Lichtenberg, T., Schaefer, L., Krissansen-Totton, J., et al. 2025, *arXiv e-prints*, arXiv:2511.16142, doi: [10.48550/arXiv.2511.16142](https://doi.org/10.48550/arXiv.2511.16142)
- Lin, Z., MacDonald, R. J., Kaltenegger, L., & Wilson, D. J. 2021, *MNRAS*, 505, 3562, doi: [10.1093/mnras/stab1486](https://doi.org/10.1093/mnras/stab1486)
- Loganathan, P., Naidu, G., & Vigneswaran, S. 2017, *Environ. Sci.: Water Res. Technol.*, 3, 37, doi: [10.1039/C6EW00268D](https://doi.org/10.1039/C6EW00268D)
- Louie, D. R., Mullens, E., Alderson, L., et al. 2025, *AJ*, 169, 86, doi: [10.3847/1538-3881/ad9688](https://doi.org/10.3847/1538-3881/ad9688)
- Luo, H., O'Rourke, J. G., & Deng, J. 2024, *Science Advances*, 10, eado7603, doi: [10.1126/sciadv.ado7603](https://doi.org/10.1126/sciadv.ado7603)
- Luque, R., Park Coy, B., Xue, Q., et al. 2024, *arXiv e-prints*, arXiv:2412.03411, doi: [10.48550/arXiv.2412.03411](https://doi.org/10.48550/arXiv.2412.03411)
- Lustig-Yaeger, J., Meadows, V. S., Crisp, D., Line, M. R., & Robinson, T. D. 2023, *PSJ*, 4, 170, doi: [10.3847/PSJ/acf3e5](https://doi.org/10.3847/PSJ/acf3e5)
- Lyu, T., Ballmer, M. D., Li, Z.-H., et al. 2025, *Nat. Commun.*, 16, 10037
- Lyu, X., & Koll, D. D. B. 2025, *arXiv e-prints*, arXiv:2510.22932. <https://arxiv.org/abs/2510.22932>
- Lyu, X., Koll, D. D. B., Cowan, N. B., et al. 2024, *ApJ*, 964, 152, doi: [10.3847/1538-4357/ad2077](https://doi.org/10.3847/1538-4357/ad2077)
- Macdonald, E. J. R., & Cowan, N. B. 2019, *MNRAS*, 489, 196, doi: [10.1093/mnras/stz2047](https://doi.org/10.1093/mnras/stz2047)
- MacDonald, R. J. 2023, *The Journal of Open Source Software*, 8, 4873, doi: [10.21105/joss.04873](https://doi.org/10.21105/joss.04873)
- MacDonald, R. J., & Madhusudhan, N. 2017, *MNRAS*, 469, 1979, doi: [10.1093/mnras/stx804](https://doi.org/10.1093/mnras/stx804)
- Malik, M., Kempton, E. M.-R., Koll, D. D. B., et al. 2019a, *ApJ*, 886, 142, doi: [10.3847/1538-4357/ab4a05](https://doi.org/10.3847/1538-4357/ab4a05)
- Malik, M., Kitzmann, D., Mendonça, J. M., et al. 2019b, *AJ*, 157, 170, doi: [10.3847/1538-3881/ab1084](https://doi.org/10.3847/1538-3881/ab1084)
- Malik, M., Grosheintz, L., Mendonça, J. M., et al. 2017, *AJ*, 153, 56, doi: [10.3847/1538-3881/153/2/56](https://doi.org/10.3847/1538-3881/153/2/56)
- Mang, J., Batalha, N. E., Morley, C. V., et al. 2026, *ApJ*, 1000, 98, doi: [10.3847/1538-4357/ae47ff](https://doi.org/10.3847/1538-4357/ae47ff)
- Mansfield, M., Kite, E. S., Hu, R., et al. 2019, *ApJ*, 886, 141, doi: [10.3847/1538-4357/ab4c90](https://doi.org/10.3847/1538-4357/ab4c90)
- Marley, M. S., Saumon, D., & Goldblatt, C. 2010, *ApJL*, 723, L117, doi: [10.1088/2041-8205/723/1/L117](https://doi.org/10.1088/2041-8205/723/1/L117)
- Martin, A. C., Emery, J. P., & Loeffler, M. J. 2022, *Icarus*, 378, 114921, doi: [10.1016/j.icarus.2022.114921](https://doi.org/10.1016/j.icarus.2022.114921)
- Mayorga, L. C., Lustig-Yaeger, J., May, E. M., et al. 2021, *PSJ*, 2, 140, doi: [10.3847/PSJ/ac0c85](https://doi.org/10.3847/PSJ/ac0c85)
- McSween, H. Y., Taylor, G. J., & Wyatt, M. B. 2009, *Science*, 324, 736, doi: [10.1126/science.1165871](https://doi.org/10.1126/science.1165871)
- Meerdink, S. K., Hook, S. J., Roberts, D. A., & Abbott, E. A. 2019, *Remote Sensing of Environment*, 230, 111196, doi: [10.1016/j.rse.2019.05.015](https://doi.org/10.1016/j.rse.2019.05.015)

- Meerdink, S. K., Roberts, D. A., King, J. Y., et al. 2016, Remote Sensing of Environment, 186, 322, doi: <https://doi.org/10.1016/j.rse.2016.08.003>
- Meier, T. G., Guimond, C. M., Pierrehumbert, R. T., et al. 2026, arXiv e-prints, arXiv:2603.02408, doi: [10.48550/arXiv.2603.02408](https://doi.org/10.48550/arXiv.2603.02408)
- Meier Valdés, E. A., Demory, B.-O., Diamond-Lowe, H., et al. 2025, A&A, 698, A68, doi: [10.1051/0004-6361/202453449](https://doi.org/10.1051/0004-6361/202453449)
- Miguel, Y., Kaltenegger, L., Fegley, B., & Schaefer, L. 2011, ApJL, 742, L19, doi: [10.1088/2041-8205/742/2/L19](https://doi.org/10.1088/2041-8205/742/2/L19)
- Milliken, R. E., Hiroi, T., Scholes, D., Slavney, S., & Arvidson, R. 2021, in LPI Contributions, Vol. 2654, Astromaterials Data Management in the Era of Sample-Return Missions Community Workshop, 2021
- Mishra, I., Lewis, N., Lunine, J., & Hand, K. P. 2025, PSJ, 6, 130, doi: [10.3847/PSJ/adcf2](https://doi.org/10.3847/PSJ/adcf2)
- Moran, S. E., Stevenson, K. B., Sing, D. K., et al. 2023, ApJL, 948, L11, doi: [10.3847/2041-8213/accb9c](https://doi.org/10.3847/2041-8213/accb9c)
- Morley, C. V., Kreidberg, L., Rustamkulov, Z., Robinson, T., & Fortney, J. J. 2017, ApJ, 850, 121, doi: [10.3847/1538-4357/aa927b](https://doi.org/10.3847/1538-4357/aa927b)
- Morley, C. V., Marley, M. S., Fortney, J. J., et al. 2014, ApJ, 787, 78, doi: [10.1088/0004-637X/787/1/78](https://doi.org/10.1088/0004-637X/787/1/78)
- Moroz, V. I. 1976, SSRv, 19, 763, doi: [10.1007/BF00173706](https://doi.org/10.1007/BF00173706)
- Morris, R. V., Vaniman, D. T., Blake, D. F., et al. 2016, Proceedings of the National Academy of Science, 113, 7071, doi: [10.1073/pnas.1607098113](https://doi.org/10.1073/pnas.1607098113)
- Mukherjee, S., Batalha, N. E., Fortney, J. J., & Marley, M. S. 2023, ApJ, 942, 71, doi: [10.3847/1538-4357/ac9f48](https://doi.org/10.3847/1538-4357/ac9f48)
- Mullen, E. K., & McCallum, I. S. 2013, Journal of Petrology, 55, 241, doi: [10.1093/petrology/egt064](https://doi.org/10.1093/petrology/egt064)
- Mullens, E., & Lewis, N. K. 2025, ApJL, 988, L43, doi: [10.3847/2041-8213/ade885](https://doi.org/10.3847/2041-8213/ade885)
- Mullens, E., Lewis, N. K., & MacDonald, R. J. 2024, ApJ, 977, 105, doi: [10.3847/1538-4357/ad8575](https://doi.org/10.3847/1538-4357/ad8575)
- Mullens, E., Schmidt, B., Kaltenegger, L., & Lewis, N. K. 2025, MNRAS, 540, 1329, doi: [10.1093/mnras/staf794](https://doi.org/10.1093/mnras/staf794)
- Mutschke, H., Posch, T., Fabian, D., & Dorschner, J. 2002, A&A, 392, 1047, doi: [10.1051/0004-6361:20021072](https://doi.org/10.1051/0004-6361:20021072)
- Nakagawa, H. 2019, in Astrobiology, ed. A. Yamagishi, T. Kakegawa, & T. Usui, 353, doi: [10.1007/978-981-13-3639-3_22](https://doi.org/10.1007/978-981-13-3639-3_22)
- National Academies of Sciences, & Medicine, E. 2021, Pathways to Discovery in Astronomy and Astrophysics for the 2020s, doi: [10.17226/26141](https://doi.org/10.17226/26141)
- Noack, L., Rivoldini, A., & Van Hoolst, T. 2017, Physics of the Earth and Planetary Interiors, 269, 40, doi: [10.1016/j.pepi.2017.05.010](https://doi.org/10.1016/j.pepi.2017.05.010)
- Oleson, K. W., Bonan, G. B., Schaaf, C., et al. 2003, Geophysical Research Letters, 30, doi: <https://doi.org/10.1029/2002GL016749>
- Palik, E. 1998, Handbook of Optical Constants of Solids III, Academic Press handbook series No. v. 3 (Elsevier Science). <https://books.google.com/books?id=nxoqxyoHfbIC>
- Palik, E. D. 1985, Handbook of optical constants of solids
- Palmer, K. F., & Williams, D. 1975, ApOpt, 14, 208, doi: [10.1364/AO.14.000208](https://doi.org/10.1364/AO.14.000208)
- Paragas, K., Knutson, H. A., Hu, R., et al. 2025, ApJ, 981, 130, doi: [10.3847/1538-4357/ada9eb](https://doi.org/10.3847/1538-4357/ada9eb)
- Patel, J. A., Brandeker, A., Kitzmann, D., et al. 2024, A&A, 690, A159, doi: [10.1051/0004-6361/202450748](https://doi.org/10.1051/0004-6361/202450748)
- Pavri, B., Head, J. W., Klose, K. B., & Wilson, L. 1992, J. Geophys. Res., 97, 13445, doi: [10.1029/92JE01162](https://doi.org/10.1029/92JE01162)
- Perez, J., Schaefer, L. K., Schwieterman, E., et al. 2025, arXiv e-prints, arXiv:2511.10801, doi: [10.48550/arXiv.2511.10801](https://doi.org/10.48550/arXiv.2511.10801)
- Philpotts, A. R., & Ague, J. J. 2022, Principles of igneous and metamorphic petrology, doi: [10.1017/9781108631419](https://doi.org/10.1017/9781108631419)
- Piaulet-Ghorayeb, C., Benneke, B., Radica, M., et al. 2024, ApJL, 974, L10, doi: [10.3847/2041-8213/ad6f00](https://doi.org/10.3847/2041-8213/ad6f00)
- Pollack, J. B., & Yung, Y. L. 1980, Annual Review of Earth and Planetary Sciences, 8, 425, doi: [10.1146/annurev.ea.08.050180.002233](https://doi.org/10.1146/annurev.ea.08.050180.002233)
- Polyansky, O. L., Kyuberis, A. A., Zobov, N. F., et al. 2018, MNRAS, 480, 2597, doi: [10.1093/mnras/sty1877](https://doi.org/10.1093/mnras/sty1877)
- Putirka, K. D., Dorn, C., Hinkel, N. R., & Unterborn, C. T. 2021, Elements, 17, 235, doi: [10.2138/gselements.17.4.235](https://doi.org/10.2138/gselements.17.4.235)
- Putirka, K. D., & Rarick, J. C. 2019, American Mineralogist, 104, 817, doi: [10.2138/am-2019-6787](https://doi.org/10.2138/am-2019-6787)
- Putirka, K. D., & Xu, S. 2021, Nature Communications, 12, 6168, doi: [10.1038/s41467-021-26403-8](https://doi.org/10.1038/s41467-021-26403-8)
- Rackham, B. V., Apai, D., & Giampapa, M. S. 2018, ApJ, 853, 122, doi: [10.3847/1538-4357/aaa08c](https://doi.org/10.3847/1538-4357/aaa08c)
- Ramirez, S. I., Coll, P., da Silva, A., et al. 2002, Icarus, 156, 515, doi: [10.1006/icar.2001.6783](https://doi.org/10.1006/icar.2001.6783)
- Reinhold, M., & Schaefer, L. 2024, in AAS/Division for Extreme Solar Systems Abstracts, Vol. 56, AASTCS10, Extreme Solar Systems V, 629.03
- Richard Drees, L., Wilding, L. P., Smeck, N. E., & Senkayi, A. L. 1989, Silica in Soils: Quartz and Disordered Silica Polymorphs (John Wiley Sons, Ltd), 913–974, doi: <https://doi.org/10.2136/sssabookser1.2ed.c19>
- Rivkin, A. S., Volquardsen, E. L., & Clark, B. E. 2006, Icarus, 185, 563, doi: [10.1016/j.icarus.2006.08.022](https://doi.org/10.1016/j.icarus.2006.08.022)
- Rodriguez, J. A. P., Domingue, D., Travis, B., et al. 2023, PSJ, 4, 219, doi: [10.3847/PSJ/acf219](https://doi.org/10.3847/PSJ/acf219)

- Ruff, S. W., & Christensen, P. R. 2007, *Geophys. Res. Lett.*, 34, L10204, doi: [10.1029/2007GL029602](https://doi.org/10.1029/2007GL029602)
- Sautter, V., Toplis, M. J., Beck, P., et al. 2016, *Lithos*, 254, 36, doi: [10.1016/j.lithos.2016.02.023](https://doi.org/10.1016/j.lithos.2016.02.023)
- Schaefer, L., & Fegley, B. 2009, *ApJL*, 703, L113, doi: [10.1088/0004-637X/703/2/L113](https://doi.org/10.1088/0004-637X/703/2/L113)
- Schwartz, J. C., & Cowan, N. B. 2015, *MNRAS*, 449, 4192, doi: [10.1093/mnras/stv470](https://doi.org/10.1093/mnras/stv470)
- Seager, S. 2010, *Exoplanet Atmospheres: Physical Processes*
- Seddio, S., Korotev, R., Jolliff, B., & Wang, A. 2015, *American Mineralogist*, 100, 1533, doi: [10.2138/am-2015-5058](https://doi.org/10.2138/am-2015-5058)
- Selsis, F., Kaltenegger, L., & Paillet, J. 2008, *Physica Scripta Volume T*, 130, 014032, doi: [10.1088/0031-8949/2008/T130/014032](https://doi.org/10.1088/0031-8949/2008/T130/014032)
- Semprich, J., Filiberto, J., Weller, M., Gorce, J., & Clark, N. 2025, *Nature Communications*, 16, doi: [10.1038/s41467-025-58324-1](https://doi.org/10.1038/s41467-025-58324-1)
- Shirley, K. A., & Glotch, T. D. 2019, *Journal of Geophysical Research (Planets)*, 124, 970, doi: [10.1029/2018JE005533](https://doi.org/10.1029/2018JE005533)
- Shirley, K. A., Glotch, T. D., Donaldson, O., et al. 2023, *Journal of Geophysical Research (Planets)*, 128, e2022JE007629, doi: [10.1029/2022JE007629](https://doi.org/10.1029/2022JE007629)
- Sparks, W. B., Parenteau, M. N., Blankenship, R. E., et al. 2021, *Astrobiology*, 21, 219, doi: [10.1089/ast.2020.2272](https://doi.org/10.1089/ast.2020.2272)
- Suto, H., Sogawa, H., Tachibana, S., et al. 2006, *MNRAS*, 370, 1599, doi: [10.1111/j.1365-2966.2006.10594.x](https://doi.org/10.1111/j.1365-2966.2006.10594.x)
- Tatsumi, Y., Sato, T., & Kodaira, S. 2015, *Earth, Planets and Space*, 67, 91, doi: [10.1186/s40623-015-0267-2](https://doi.org/10.1186/s40623-015-0267-2)
- Taylor, F. W., Svedhem, H., & Head, J. W. 2018, *SSRv*, 214, 35, doi: [10.1007/s11214-018-0467-8](https://doi.org/10.1007/s11214-018-0467-8)
- Tenthoff, M., Wohlfarth, K., Wöhler, C., Zieba, S., & Kreidberg, L. 2024, in *European Planetary Science Congress, EPSC2024*–649, doi: [10.5194/epsc2024-649](https://doi.org/10.5194/epsc2024-649)
- Teske, J. K., Wallack, N. L., Piette, A. A. A., et al. 2025, *arXiv e-prints*, arXiv:2509.17231, doi: [10.48550/arXiv.2509.17231](https://doi.org/10.48550/arXiv.2509.17231)
- Thompson, J. O., Williams, D. B., Lee, R. J., & Ramsey, M. S. 2021, *Journal of Geophysical Research (Solid Earth)*, 126, e2021JB022157, doi: [10.1029/2021JB022157](https://doi.org/10.1029/2021JB022157)
- Tikoo, S. M., & Elkins-Tanton, L. T. 2017, *Philosophical Transactions of the Royal Society of London Series A*, 375, 20150394, doi: [10.1098/rsta.2015.0394](https://doi.org/10.1098/rsta.2015.0394)
- Toon, O. B., McKay, C. P., Ackerman, T. P., & Santhanam, K. 1989, *J. Geophys. Res.*, 94, 16287, doi: [10.1029/JD094iD13p16287](https://doi.org/10.1029/JD094iD13p16287)
- Triaud, A. H. 2005, *Database of Optical Constants for Cosmic Dust*, unpublished
- Trumbo, S. K., Brown, M. E., & Hand, K. P. 2019, *Science Advances*, 5, aaw7123, doi: [10.1126/sciadv.aaw7123](https://doi.org/10.1126/sciadv.aaw7123)
- Tsiaras, A., Rocchetto, M., Waldmann, I. P., et al. 2016, *ApJ*, 820, 99, doi: [10.3847/0004-637X/820/2/99](https://doi.org/10.3847/0004-637X/820/2/99)
- Tusay, N., Wright, J. T., Beatty, T. G., et al. 2025, *ApJL*, 987, L6, doi: [10.3847/2041-8213/addfd0](https://doi.org/10.3847/2041-8213/addfd0)
- Udry, A., Gazel, E., & McSween, H. Y. 2018, *Journal of Geophysical Research (Planets)*, 123, 1525, doi: [10.1029/2018JE005602](https://doi.org/10.1029/2018JE005602)
- Underwood, D. S., Tennyson, J., Yurchenko, S. N., et al. 2016, *MNRAS*, 459, 3890, doi: [10.1093/mnras/stw849](https://doi.org/10.1093/mnras/stw849)
- Unterborn, C. T., & Panero, W. R. 2017, *ApJ*, 845, 61, doi: [10.3847/1538-4357/aa7f79](https://doi.org/10.3847/1538-4357/aa7f79)
- Vaniman, D. T., Bish, D. L., Chipera, S. J., et al. 2004, *Nature*, 431, 663, doi: [10.1038/nature02973](https://doi.org/10.1038/nature02973)
- Vaniman, D. T., Bish, D. L., Ming, D. W., et al. 2014, *Science*, 343, 1243480, doi: [10.1126/science.1243480](https://doi.org/10.1126/science.1243480)
- Vaughan, D. 2014, *Minerals: A Very Short Introduction* (Oxford University Press), doi: [10.1093/actrade/9780199682843.001.0001](https://doi.org/10.1093/actrade/9780199682843.001.0001)
- Vaughan, R. G., Hook, S. J., Calvin, W. M., & Taranik, J. V. 2005, *Remote Sensing of Environment*, 99, 140, doi: <https://doi.org/10.1016/j.rse.2005.04.030>
- Veras, D., & Wolszczan, A. 2019, *MNRAS*, 488, 153, doi: [10.1093/mnras/stz1721](https://doi.org/10.1093/mnras/stz1721)
- Villanueva, G. L., Smith, M. D., Protopapa, S., Faggi, S., & Mandell, A. M. 2018, *JQSRT*, 217, 86, doi: [10.1016/j.jqsrt.2018.05.023](https://doi.org/10.1016/j.jqsrt.2018.05.023)
- Wachiraphan, P., Berta-Thompson, Z. K., Diamond-Lowe, H., et al. 2024, *arXiv e-prints*, arXiv:2410.10987, doi: [10.48550/arXiv.2410.10987](https://doi.org/10.48550/arXiv.2410.10987)
- Wakeford, H. R., Mayorga, L. C., Barstow, J. K., et al. 2025, *arXiv e-prints*, arXiv:2506.22839, doi: [10.48550/arXiv.2506.22839](https://doi.org/10.48550/arXiv.2506.22839)
- Wang, H. S., Liu, F., Ireland, T. R., et al. 2019, *MNRAS*, 482, 2222, doi: [10.1093/mnras/sty2749](https://doi.org/10.1093/mnras/sty2749)
- Wang, H. S., Quanz, S. P., Yong, D., et al. 2022, *MNRAS*, 513, 5829, doi: [10.1093/mnras/stac1119](https://doi.org/10.1093/mnras/stac1119)
- Wang, R., MacDonald, R. J., Gibson, N. P., & Lewis, N. K. 2025, *AJ*, 169, 328, doi: [10.3847/1538-3881/adcac7](https://doi.org/10.3847/1538-3881/adcac7)
- Wang, Z., Zeng, X., Barlage, M., et al. 2004, *Journal of Hydrometeorology - J HYDROMETEOROL*, 5, doi: [10.1175/1525-7541\(2004\)005<0003:UMBAAD>2.0.CO;2](https://doi.org/10.1175/1525-7541(2004)005<0003:UMBAAD>2.0.CO;2)
- Warren, S. G. 1984, *ApOpt*, 23, 1206, doi: [10.1364/AO.23.001206](https://doi.org/10.1364/AO.23.001206)
- Weiner Mansfield, M., Xue, Q., Zhang, M., et al. 2024, *ApJL*, 975, L22, doi: [10.3847/2041-8213/ad8161](https://doi.org/10.3847/2041-8213/ad8161)
- Whittaker, E. A., Malik, M., Ih, J., et al. 2022, *AJ*, 164, 258, doi: [10.3847/1538-3881/ac9ab3](https://doi.org/10.3847/1538-3881/ac9ab3)

- Winter, J. 2014, *Principles of Igneous and Metamorphic Petrology*, 2nd edn. (Pearson)
- Wordsworth, R., & Kreidberg, L. 2022, *ARA&A*, 60, 159, doi: [10.1146/annurev-astro-052920-125632](https://doi.org/10.1146/annurev-astro-052920-125632)
- Wurz, P., Jäggi, N., Galli, A., et al. 2025, *PSJ*, 6, 24, doi: [10.3847/PSJ/ad95fa](https://doi.org/10.3847/PSJ/ad95fa)
- Wyatt, M. B., Hamilton, V. E., McSween Jr., H. Y., Christensen, P. R., & Taylor, L. A. 2001, *Journal of Geophysical Research: Planets*, 106, 14711, doi: <https://doi.org/10.1029/2000JE001356>
- Xu, S., & Bonsor, A. 2021, *Elements*, 17, 241, doi: [10.2138/gselements.17.4.241](https://doi.org/10.2138/gselements.17.4.241)
- Xue, Q., Bean, J. L., Zhang, M., et al. 2024, *ApJL*, 973, L8, doi: [10.3847/2041-8213/ad72e9](https://doi.org/10.3847/2041-8213/ad72e9)
- Xue, Q., Zhang, M., Coy, B. P., et al. 2025, *ApJL*, 995, L52, doi: [10.3847/2041-8213/ae2098](https://doi.org/10.3847/2041-8213/ae2098)
- Yen, A. S., Morris, R. V., Ming, D. W., et al. 2021, *Journal of Geophysical Research (Planets)*, 126, e06569, doi: [10.1029/2020JE006569](https://doi.org/10.1029/2020JE006569)
- Yurchenko, S. N., Mellor, T. M., Freedman, R. S., & Tennyson, J. 2020, *MNRAS*, 496, 5282, doi: [10.1093/mnras/staa1874](https://doi.org/10.1093/mnras/staa1874)
- Yurchenko, S. N., Owens, A., Kefala, K., & Tennyson, J. 2024, *MNRAS*, 528, 3719, doi: [10.1093/mnras/stae148](https://doi.org/10.1093/mnras/stae148)
- Zalasiewicz, J. 2016, *Rocks: A Very Short Introduction*, doi: [10.1093/actrade/9780198725190.001.0001](https://doi.org/10.1093/actrade/9780198725190.001.0001)
- Zeidler, S., Posch, T., & Mutschke, H. 2013, *A&A*, 553, A81, doi: [10.1051/0004-6361/201220459](https://doi.org/10.1051/0004-6361/201220459)
- Zelakiewicz, A. S., Mullens, E., Kaltenegger, L., & Savransky, D. 2026, arXiv e-prints, arXiv:2603.25694, doi: [10.48550/arXiv.2603.25694](https://doi.org/10.48550/arXiv.2603.25694)
- Zeng, X., Jin, Z., Dong, C., et al. 2025, *Geophys. Res. Lett.*, 52, e2025GL116170, doi: [10.1029/2025GL116170](https://doi.org/10.1029/2025GL116170)
- Zhang, M., Hu, R., Inglis, J., et al. 2024, *ApJL*, 961, L44, doi: [10.3847/2041-8213/ad1a07](https://doi.org/10.3847/2041-8213/ad1a07)
- Zieba, S., Kreidberg, L., Ducrot, E., et al. 2023, *Nature*, 620, 746, doi: [10.1038/s41586-023-06232-z](https://doi.org/10.1038/s41586-023-06232-z)